

THE IRRATIONALITY OF THE 5-ADIC ZETA VALUE $\zeta_5(3)$

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ABSTRACT. We prove that the Kubota–Leopoldt 5-adic zeta value $\zeta_5(3) \in \mathbb{Q}_5$ is irrational. The proof is the arithmetic holonomy method of Calegari, Dimitrov and Tang, in the form they used for $\zeta_2(5)$: Calegari’s p -adic Apéry construction on $X_0(5)$ gives an overconvergent function H which is analytic on $|x|_5 < 5^3$ and whose coefficients would be rational with denominators $[1, \dots, n]^3$ if $\zeta_5(3)$ were rational. For every fixed $\alpha < 3$ this analyticity gives the coefficient bound required by the adelic holonomy argument with effective radius $R_5 = 5^\alpha$; we take $\alpha = 2.995$. A holonomy bound with an archimedean template chosen among *all* holomorphic self-maps of the q -disc then caps the dimension of the space such functions span below the four exhibited. The multi-place form of the bound, allowing real exponents $R_p = p^{\alpha_p}$, is obtained from the published Bost–Charles form by a floor-twist of the integral structure in Bost’s slopes inequality. Calegari’s original criterion fails for this value by 11%; the crude sup-norm holonomy bound fails; the refined bound succeeds with a certified positive margin.

1. STATEMENT

For a prime p , let ζ_p denote the Kubota–Leopoldt p -adic zeta function and, for $k \geq 1$, $\zeta_p(2k+1) \in \mathbb{Q}_p$ its value at the positive odd integer $2k+1$ (interpolating $\zeta^*(s) = (1-p^{-s})\zeta(s)$ at negative integers of the right parity). Calegari [5] proved $\zeta_p(3) \notin \mathbb{Q}$ for $p = 2, 3$ and $L_2(2, \chi_{-4}) \notin \mathbb{Q}$; Calegari–Dimitrov–Tang [7] proved $\zeta_2(5) \notin \mathbb{Q}$ (see also Lai–Sprang–Zudilin [12] for an Apéry-type proof); Beukers [1] proved $\zeta_p(2), \zeta_p(3) \notin \mathbb{Q}$ for $p = 2, 3$ and obtained, for $p = 5$, only that $\zeta_5(3) \pm L_5(3, \chi_5)$ is irrational; Lai–Lupu–Sprang [11] proved that for every $p \geq 5$ some $\zeta_p(i)$ with odd $i \in [3, p + p/\log p + 5]$ is irrational.

Theorem 1.1. $\zeta_5(3) \notin \mathbb{Q}$.

The characterisation of $\zeta_5(3)$ used throughout is the one of [5, §2.2], [7, §6.2]: $\zeta_5(3)/2$ is the unique element $\eta \in \mathbb{Q}_5$ such that

$$(1) \quad E_{-2}^* := \eta + \sum_{n \geq 1} \left(\sum_{d|n, 5 \nmid d} d^{-3} \right) q^n$$

is the q -expansion of an overconvergent 5-adic modular form of weight -2 and level $\Gamma_0(5)$.

2. THE METHOD

2.1. Calegari’s construction. Let $x = (\Delta(5\tau)/\Delta(\tau))^{1/4} = q \prod_{n \geq 1} ((1 - q^{5n})/(1 - q^n))^6 \in q + q^2\mathbb{Z}[[q]]$, a Hauptmodul of $X_0(5)$ with $x = 0$ at the cusp ∞ and $x = \infty$ at the cusp 0 ; its formal inverse lies in $q + x + x^2\mathbb{Z}[[x]]$. Let $E_2^* = E_2(\tau) - 5E_2(5\tau) \in 1 + q\mathbb{Z}[[q]]$ (here $4k/B_{2k} = 24$ makes E_2 integral), and set

$$H := E_2^* \cdot E_{-2}^* = \sum_{n \geq 0} h_n x^n, \quad h_n = a_n + \eta b_n, \quad a_n, b_n \in \mathbb{Q}.$$

H is a weight-0 overconvergent form, i.e. a function on the rigid curve $X_0(5)$. The Fricke involution sends $x \mapsto 5^{-3}/x$ and exchanges the two components $\{|x|_5 \leq 1\}$, $\{|x|_5 \geq 5^3\}$ of the

ordinary locus; by Buzzard's theorem [4, Thm. 5.2] the slope-0 form E_{-2}^* extends over the whole supersingular annulus, so $H(x)$ converges on $|x|_5 < 5^3$. We use only this open-disc continuation, not any assertion that the radius is exactly 5^3 ; see Proposition 4.3.

Suppose $\zeta_5(3) \in \mathbb{Q}$. Then $H \in \mathbb{Q}[[x]]$.

2.2. The holonomy bound. We use the notation of [6, Thm. 2.5.1]: m functions $f_1, \dots, f_m \in \mathbb{Q}[[x]]$, $\mathbb{Q}(x)$ -linearly independent, with denominator types encoded by a step-shaped array $\mathbf{b} = (b_{i,j})$ (row sums σ_i , u_j the number of zero entries in column j) and

$$\tau(\mathbf{b}) = \frac{1}{m^2} \sum_{i=1}^m (2i-1)\sigma_i;$$

a holomorphic $\varphi : (\overline{\mathbb{D}}, 0) \rightarrow (\mathbb{C}, 0)$ such that each $f_i \circ \varphi$ is meromorphic on $\mathbb{D}$; and the Bost–Charles integral $\text{BC}(\varphi) = \iint_{\mathbb{T}^2} \log |\varphi(z) - \varphi(w)| d\mu(z) d\mu(w)$. Their theorem: if $\log |\varphi'(0)| > \tau(\mathbf{b})$ and the f_i are holonomic, then $m \leq \text{BC}(\varphi) / (\log |\varphi'(0)| - \tau(\mathbf{b}))$.

For a p -adic target one needs the version with p -adic radii, used in [7, §6.2] and attributed there to a forthcoming paper:

Theorem 2.1 (multi-place bound). *In the situation above, let S be a finite set of primes and for $p \in S$ let $R_p = p^{\alpha_p}$ with $\alpha_p > 0$ real, such that the coefficients $c_{i,n}$ of x^n in f_i satisfy $|c_{i,n}|_p \leq Cn^A R_p^{-n}$ for all i, n . If $\log |\varphi'(0)| + (1 - \frac{1}{m}) \sum_{p \in S} \log R_p > \tau(\mathbf{b})$ then*

$$m \leq \frac{\text{BC}(\varphi) + \sum_{p \in S} \log R_p}{\log |\varphi'(0)| + \sum_{p \in S} \log R_p - \tau(\mathbf{b})}.$$

A proof from published ingredients is given in §3. The crude (sup-norm) analogue, with $\sup_{\mathbb{T}} \log |\varphi|$ in place of BC , is Dimitrov's arithmetic holonomicity theorem [8, 9]; it does not suffice here (Remark 6.2).

2.3. The inventory and the reduction to one inequality. By Kontsevich–Zagier [10, §2.3], a modular form of weight w expressed in a modular function satisfies a linear ODE of minimal order $w+1$; for the product H this gives an *inhomogeneous* ODE over $\mathbb{Q}(x)$ of minimal order 3, so that $\{1, H, \delta H, \delta^2 H\}$, $\delta = x d/dx$, are $m = 4$ $\mathbb{Q}(x)$ -linearly independent functions (Lemma 4.2). Their denominators are of type $[1, \dots, n]^3$ (Proposition 4.1), so $\mathbf{b}$ has $r = 3$, $b_{i,j} = 1$ for $i \geq 2$, $u_j = 1$, and $\tau(\mathbf{b}) = 3(1 - \frac{1}{16}) = \frac{45}{16}$. Fix

$$\alpha = 2.995 < 3, \quad R_5 := 5^\alpha, \quad L := \log R_5 = 2.995 \log 5,$$

which is permitted by Proposition 4.3, and put $\rho := |\varphi'(0)|$. Theorem 2.1 contradicts $m = 4$ exactly when

$$(2) \quad \text{cost}(\varphi) := \text{BC}(\varphi) - 4 \log \rho < 3(L - 4 + \tfrac{1}{4}) = 3.2107997\dots =: \text{budget}.$$

The budget is arithmetic ($p = 5$, $k = 1$, together with the chosen $\alpha < 3$); the cost depends only on the template.

2.4. Admissible templates. $H(x(q))$ and its δ -derivatives are meromorphic on the whole disc $|q| < 1$ (E_2^* is holomorphic on $\mathfrak{H}$, the coefficients of E_{-2}^* are bounded, and d/dx only introduces poles at the critical points of x). Hence $\varphi = x \circ \psi$ is admissible for *every* holomorphic $\psi : \mathbb{D} \rightarrow \mathbb{D}$ with $\psi(0) = 0$ — no injectivity is required and $\psi(\mathbb{D})$ need not be a Jordan domain. Writing $\psi(z) = z \exp(c_0 + \sum_{k \geq 1} c_k z^k)$ with real c_k , the only constraint is $\max_{|z|=1} \log |\psi| \leq 0$, and $\log \rho = c_0$ exactly (since $x'(0) = 1$). We optimise (2) over this family. The points of $|q| = 1$ at which x is unbounded are the $\Gamma_0(5)$ -images of the cusp 0, which are dense on the circle (unlike $p = 2$, where CDT's contour may touch $q = -1$); the optimum stays inside $|q| \leq 0.8363$ with the constraint inactive.

3. PROOF OF THEOREM 2.1

We follow the proof of [6, Thm. *main:BC form*, §6.3], which runs Bost's slopes inequality [2] on the filtered evaluation module of auxiliary polynomials, and change only the integral structure at the primes of S . Recall their setup: E_D is the lattice of $\sum_{i=1}^m Q_i f_i$ with $Q_i \in \mathbb{Z}[1/x]_{\leq D}$ (weighted by the lcm factors prescribed by $\mathbf{b}$), filtered by vanishing order n , with the evaluation maps $\psi_D^{(n)} : E_D^{(n)} / E_D^{(n+1)} \rightarrow G^{(n)} = \mathbb{Z} \cdot x^{n-D}$ onto the leading coefficient; the archimedean norm on E_D is the Bost–Charles metric attached to φ , and $\|x^{n-D}\|_\infty = 1$. Bost's inequality reads

$$(3) \quad \widehat{\deg E_D} \leq \sum_{n \in \mathcal{V}_D} \left[\widehat{\mu}_{\max}(\overline{G^{(n)}}) + h_\infty(\psi_D^{(n)}) + h_{\text{fin}}(\psi_D^{(n)}) \right],$$

where $\mathcal{V}_D$ is the set of attained vanishing orders, $\#\mathcal{V}_D = m(D+1)$ and $\sum_{n \in \mathcal{V}_D} n = \frac{m^2}{2} D^2 + o(D^2)$ by their Kolchin-type lemma. CDT evaluate every term: $\widehat{\deg E_D} = (\frac{m}{2} \text{BC}(\varphi) + \dots) D^2 + o(D^2)$, $\sum h_\infty \leq (-\frac{m^2}{2} \log |\varphi'(0)| + m \text{BC}(\varphi)) D^2 + o(D^2)$, and $\sum h_{\text{fin}}$ accounts for the lcm denominators and produces $\tau(\mathbf{b})$.

The floor-twist. Put $a_{p,j} = \lfloor \alpha_p j \rfloor$. Replace the two lattices by

$$\Lambda_D^{\text{ad}} := \bigoplus_{j=0}^D \mathbb{Z} \left(\prod_{p \in S} p^{-a_{p,j}} \right) x^{-j} \supset \mathbb{Z}[1/x]_{\leq D}, \quad G_{\text{ad}}^{(n)} := \left(\prod_{p \in S} p^{\lfloor \alpha_p (n-D) \rfloor} \right) \mathbb{Z} \cdot x^{n-D}.$$

Use Λ_D^{ad} in place of $\mathbb{Z}[1/x]_{\leq D}$ in each of the m summands of E_D , keeping every archimedean norm unchanged. Since $E_{D,\mathbb{R}}$ and $\text{Gr}_n(F)_{\mathbb{R}}$ and their Euclidean norms are unchanged and h_v depends only on the v -adic norms, all archimedean estimates of [6] hold verbatim; at primes $\ell \notin S$ nothing changes. The floor errors are $O(1)$ per basis vector or graded piece and therefore contribute only $O(D)$ or $O(D \log D)$, hence $o(D^2)$, to the slopes inequality. Three computations account for the twist.

(1) Since

$$\sum_{j=0}^D \lfloor \alpha_p j \rfloor = \frac{\alpha_p}{2} D(D+1) + O(D),$$

the enlargement $\Lambda_D^{\text{ad}} \supset \mathbb{Z}[1/x]_{\leq D}$ changes the Arakelov degree by

$$\widehat{\deg E_D^{\text{ad}}} = \widehat{\deg E_D} + \frac{m}{2} D^2 \sum_{p \in S} \alpha_p \log p + o(D^2).$$

(2) The rank-one lattice $\overline{G_{\text{ad}}^{(n)}}$ has a generator of archimedean norm $\prod_p p^{\lfloor \alpha_p (n-D) \rfloor}$, hence

$$\widehat{\mu}_{\max}(\overline{G_{\text{ad}}^{(n)}}) = - \sum_{p \in S} \lfloor \alpha_p (n-D) \rfloor \log p.$$

Using $\sum_{n \in \mathcal{V}_D} n = \frac{m^2}{2} D^2 + o(D^2)$ and $\#\mathcal{V}_D = m(D+1)$, summation gives

$$- \left(\frac{m^2}{2} - m \right) D^2 \sum_{p \in S} \alpha_p \log p + o(D^2).$$

(3)

Lemma 3.1 (p -adic operator norm). *For $p \in S$ and every n , $h_p(\psi_D^{(n)}) \leq \log(Cn^A)$.*

Proof. At p , the norm on E_D^{ad} is

$$\left\| \sum_{i,j} q_{ij} f_i x^{-j} \right\|_{E,p} = \max_{i,j} |q_{ij}|_p p^{-\lfloor \alpha_p j \rfloor},$$

up to the p -parts of the lcm factors, which contribute only $o(D^2)$ in CDT's finite-place estimate, while

$$\|c x^{n-D}\|_{G,p} = |c|_p p^{\lfloor \alpha_p(n-D) \rfloor}.$$

Writing $k = n - D$, the leading coefficient of $Q = \sum q_{ij} f_i x^{-j}$ is $c_n(Q) = \sum_{i,j} q_{ij} c_{i,k+j}$. By the ultrametric inequality and the hypothesis on $c_{i,r}$,

$$|c_n(Q)|_p \leq \max_{i,j} |q_{ij}|_p C n^A p^{-\alpha_p(k+j)} \leq C n^A p^{-\lfloor \alpha_p k \rfloor} \max_{i,j} |q_{ij}|_p p^{-\lfloor \alpha_p j \rfloor},$$

because $\lfloor \alpha_p k \rfloor + \lfloor \alpha_p j \rfloor \leq \alpha_p(k+j)$. Thus $\|\psi_D^{(n)}[Q]\|_{G,p} \leq C n^A \|Q\|_{E,p}$, and the same inequality holds for the quotient norm. $\square$

Summed over $\mathcal{V}_D$, (3) is $O(D \log D)$. Inserting (1)–(3) into (3) and dividing by D^2 adds $\frac{m}{2}\Lambda$ to the left and $-(\frac{m^2}{2} - m)\Lambda$ to the right of CDT's inequality, where

$$\Lambda = \sum_{p \in S} \alpha_p \log p = \sum_{p \in S} \log R_p.$$

After their optimisation over the auxiliary parameters this is

$$m(\log |\varphi'(0)| - \tau(\mathbf{b})) + (m-1)\Lambda \leq \text{BC}(\varphi),$$

which is the statement. $\square$

Remark 3.2. With $m = 2$, $\mathbf{b} = 0$ and φ univalent, Theorem 2.1 reduces to the Borel–Dwork criterion $\rho \prod_p R_p > 1$; the weight $1 - \frac{1}{m}$ on Λ is forced by the product formula (the margin is invariant under $x \mapsto x/p^v$). The high-dimensional proof of the rearrangement-numerator form [6, Thm. *elementary form*] can be twisted in the same way but, as far as we can see, only yields the weight $(1 - \frac{1}{m})^2$; we therefore use the Bost–Charles numerator throughout.

4. THE ARITHMETIC INPUTS

Proposition 4.1 (denominator type). *Write $H = \sum h_n x^n$ with $\zeta_5(3) \in \mathbb{Q}$ assumed, and let D'_n be the least common multiple of the integers $\leq n$ prime to 5. Then $(D'_n)^3 h_n \in \mathbb{Z}[1/5]$ after clearing a fixed denominator, and the exponent 3 is sharp. The same holds for $\delta H, \delta^2 H$ (the coefficients are $n^j h_n$).*

Proof. $E_2^* \in 1 + q\mathbb{Z}[[q]]$; the q -coefficients of E_{-2}^* are $\sum_{d|n, 5 \nmid d} d^{-3}$, whose denominators divide $(D'_n)^3$; and $q \in x + x^2\mathbb{Z}[[x]]$. Hence $(D'_n)^3 \cdot [x^n](E_2^* E_{-2}^*) \in \mathbb{Z}[1/5]$. Sharpness was verified exactly for $n \leq 60$. The use of δ rather than d/dx keeps $\delta^j H$ literally within the shape required by [6], while spanning the same $\mathbb{Q}(x)$ -module as $\{H, H', H''\}$. $\square$

The prime-to-5 lcm grows like e^n (prime number theorem), so the array is as stated in §2.3; the 5-power content of h_n is what R_5 measures.

Lemma 4.2 (independence). *$1, H, \delta H, \delta^2 H$ are $\mathbb{Q}(x)$ -linearly independent.*

Proof. A relation $a_0 + a_1 H + a_2 \delta H + a_3 \delta^2 H = 0$ with $a_i \in \mathbb{Q}(x)$ not all zero and $(a_1, a_2, a_3) \neq 0$ is an inhomogeneous linear ODE for H of order ≤ 2 , contradicting the minimal order 3 of Kontsevich–Zagier (structurally: $\theta^3 E'_{-2} = E_4 - E_4(5\tau)$, the Eichler cocycle has degree ≤ 2 , and E_2^* is a period of the Sym^2 of the Picard–Fuchs system); $(a_1, a_2, a_3) = 0$ forces $a_0 = 0$. The

rank was also verified directly: no relation with polynomial coefficients of degree ≤ 120 exists (computed modulo $2^{61} - 1$). $\square$

Proposition 4.3 (5-adic overconvergence). *The series $H(x) = \sum_{n \geq 0} h_n x^n$ is analytic for $|x|_5 < 5^3$. Consequently, for every real $\alpha < 3$ there is a constant C_α such that*

$$|h_n|_5 \leq C_\alpha 5^{-\alpha n} \quad (n \geq 0).$$

The same estimate holds for the coefficients of δH and $\delta^2 H$.

Proof. Buzzard [4, Thm. 5.2] (slope 0, Eisenstein family) gives analyticity of E_{-2}^* on the ordinary component containing ∞ together with the entire supersingular annulus, which in the coordinate x is the disc $|x|_5 < 5^3$; cf. [5, §3] for $p = 2, 3$ and [7, §6.2]. Since E_2^* is classical, the product $H = E_2^* E_{-2}^*$ is analytic on the same disc.

Fix $\alpha < 3$. Because $5^\alpha < 5^3$, convergence on the closed disc $|x|_5 \leq 5^\alpha$ implies that the sequence $|h_n|_5 5^{\alpha n}$ is bounded; this gives the displayed estimate. The coefficient of x^n in $\delta^j H$ is $n^j h_n$, and $|n|_5 \leq 1$, so the same bound holds for $j = 1, 2$.

Exact computations for $n \leq 2000$ are consistent with the stronger asymptotic behaviour $v_5(h_n) = 3n + O(\log n)$, but a finite computation does not prove that the radius is exactly 5^3 , and exactness is not used here. $\square$

5. THE TEMPLATE AND THE NUMERICAL CERTIFICATE

Take $\psi(z) = z \exp(\sum_{k=0}^{40} c_k z^k)$ with

$$\begin{aligned} c_{0..5} &= (-0.531289158, -0.225379074, -0.127598877, -0.001824369, -0.027735780, -0.041418343) \\ c_{6..11} &= (+0.051444137, +0.026087140, -0.010853938, +0.001314954, -0.003547849, -0.023294126) \\ c_{12..17} &= (+0.000125751, +0.020158067, -0.001454440, -0.004262361, +0.007019143, -0.000198726) \\ c_{18..23} &= (-0.007684408, -0.006339902, +0.005397468, +0.001260337, +0.000134428, +0.014051878) \\ c_{24..29} &= (-0.011391873, -0.005386029, -0.000202917, +0.001464054, -0.000123972, +0.008670335) \\ c_{30..35} &= (-0.000051833, -0.006877094, -0.001450246, -0.001060183, +0.002930635, +0.002103211) \\ c_{36..40} &= (+0.001750273, -0.002516914, -0.002244428, +0.000524421, +0.001281738). \end{aligned}$$

Admissibility. On $|z| = 1$, $\log |\psi| = u(t) = c_0 + \sum c_k \cos 2\pi kt$ has Lipschitz constant $2\pi \sum k|c_k| = 2\pi \cdot 4.3589$; sampling u on 2^{22} points and adding the Lipschitz correction gives $\max u \leq -0.1788 < 0$, so $\psi(\mathbb{D}) \subset \mathbb{D}$ by the maximum principle. *Conformal size.* $\log \rho = c_0 = -0.531289158$ exactly. *The integrals.* With $g(t) = \log |x(\psi(e^{2\pi it}))|$, the a-priori bound $|g'| \leq 2\pi \max |\Lambda|(1 + \sum k|c_k|)$, $\Lambda = q d \log x / dq = -\frac{1}{4}(E_2(q) - 5E_2(q^5))$, and the triangle-inequality majorant of the Lambert series on $|q| \leq 0.836266$, give $\text{Lip}(g) \leq 11427$; the rearrangement integral $\text{RE} = \int_0^1 2t g^*(t) dt$ on $N = 2^{24}$ nodes is then certified to $\pm \text{Lip}/N = \pm 6.8 \cdot 10^{-4}$. The Bost–Charles integral is computed on the same grid and is N -stable to 10^{-8} from 2^{12} to 2^{24} nodes (its convergence is $O(1/N)$ since φ is not injective on the boundary; we have no interval-arithmetic bound for it, and record both).

quantity	value	provenance
m	4	Lemma 4.2
$\tau(\mathbf{b})$	$45/16 = 2.8125$	Prop. 4.1
$\log R_5$	$2.995 \log 5 = 4.820266548$	Prop. 4.3
$\log \rho = c_0$	-0.531289158	exact
$\max q $ on contour	0.836265	
$\text{RE}(\varphi)$	1.066965 ± 0.00068	$N = 2^{24}$, a-priori error
$\text{BC}(\varphi)$	1.0355293	N -converged to 10^{-8}
bound (RE)	≤ 3.98782	certified < 4 using the $+6.8 \cdot 10^{-4}$ error
bound (BC)	3.9660586	numerical cross-check only
crude sup-norm form	5.714	fails (Remark 6.2)

Proof of Theorem 1.1. Assume $\zeta_5(3) \in \mathbb{Q}$. By §2 and Proposition 4.3, $\{1, H, \delta H, \delta^2 H\}$ satisfy the hypotheses of Theorem 2.1 with $S = \{5\}$, $R_5 = 5^{2.995}$, $\mathbf{b}$ as in §2.3 and $\varphi = x \circ \psi$. The entry condition is

$$c_0 + \frac{3}{4} \cdot 2.995 \log 5 = 3.0839107 \dots > \frac{45}{16}.$$

Moreover Nazarov’s comparison inequality [6, Prop. 8.1.13] and the certified rearrangement computation give

$$\text{BC}(\varphi) \leq \text{RE}(\varphi) \leq 1.066965 + 0.00068 = 1.067645.$$

Hence Theorem 2.1 gives

$$4 \leq \frac{1.067645 + 2.995 \log 5}{c_0 + 2.995 \log 5 - 45/16} < 3.98782 < 4,$$

a contradiction. $\square$

Remark 5.1 (effective measure). An earlier draft also stated the effective irrationality measure 390. That numerical exponent used the boundary choice $R_5 = 5^3$ inside the quantitative version of the holonomy argument. The present proof needs only the qualitative multi-place theorem and the floor-twist above; we therefore do not record an effective irrationality measure here. A quantitative measure can be reinstated once the quantitative floor-twist, including its lower-order terms, is checked separately.

6. REMARKS

Remark 6.1 (Calegari’s criterion). Calegari’s p -adic Apéry argument [5] compares the p -adic smallness of $b_n \eta - a_n$ with the archimedean size of the cleared coefficients; in his normalisation the criterion is $\theta_p > 1$ with $\theta_p = \sigma_p \log p / (k + \log \lambda_1)$, and for $(p, k) = (5, 1)$ one finds $\theta_5 = 0.8918$, his own recorded failure. The holonomy bound replaces the naive archimedean disc of radius $1/\lambda_1 = 5^{-3/2}$ by the template φ , whose conformal size $e^{c_0} = 0.588$ is 6.6 times larger, at the price of the numerator $\text{BC}(\varphi)$.

Remark 6.2 (the crude form fails). Dimitrov’s sup-norm bound

$$m \leq \frac{\sup_{\mathbb{T}} \log |\varphi| + \sum \log R_p}{\log |\varphi'(0)| + \sum \log R_p - \tau},$$

optimised over discs and lune-bitten discs in the q -disc, gives at best $m \leq 5.71$ (cost 4.035 against budget 2.485), so it does not prove the theorem; the Bost–Charles numerator is essential. By contrast for CDT’s $\zeta_2(5)$ the crude form already suffices ($5.56 < 6$ in [9]).

Remark 6.3 (sensitivity). The proof deliberately uses $R_5 = 5^{2.995}$ rather than asserting an exact radius. For the fixed template, including the stated $6.8 \cdot 10^{-4}$ numerical error, the certified rearrangement numerator requires only $\alpha > 2.99128\dots$, so there is room to stay strictly inside the open disc $|x|_5 < 5^3$. The argument does not survive a change of the weight $1 - \frac{1}{m}$ in Theorem 2.1 to $(1 - \frac{1}{m})^2$; that weight remains the load-bearing step, and §3 is written so that it can be checked against [6, §6.3] line by line.

Remark 6.4 (other values). The same computation gives, for (p, k) with $p - 1 \mid 24$: $\zeta_7(3)$ fails by 2.05 nats, $\zeta_3(5)$ by 2.37, $\zeta_2(7)$ by 6.46, and $\zeta_5(5)$ fails the entry condition for every template $(3 \log 5 \cdot \frac{5}{6} < \frac{35}{36} \cdot 5)$. It reproduces the known cases: Calegari's $\zeta_2(3), \zeta_3(3), L_2(2, \chi_{-4})$ with large margins, CDT's $\zeta_2(5)$ with their printed numbers (4.43206 and 4.48866 for their two contours, improved to 4.4389 here), and $L_3(2, \chi_{-3})$ on $X_0(3)$ (Beukers).

Remark 6.5 (novelty). We have not found $\zeta_5(3) \notin \mathbb{Q}$ in the literature cited above; in particular [1] reaches only $\zeta_5(3) \pm L_5(3, \chi_5)$ and [11] only the existence of an irrational value in a range. This draft is circulated for that check.

Reproducibility. All scripts and logs are in `lattice/padic_holonomy/` of the project repository; the full ledger of what is exact, numerically converged, and assumed is `consolidation/PADIC_HOLONOMY_CENSUS.md`, and the proof of Theorem 2.1 with its checks is `consolidation/ADELIC_HOLONOMY.md` §2.6.

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